

AIM: Key Exchange using Diffie-Hellman

Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE:

```
print("=" * 50)
print("    DIFFIE-HELLMAN KEY EXCHANGE")
print("=" * 50)
```

```
p = int(input("Enter a prime number (p): "))
g = int(input("Enter a generator (g): "))
```

```
print("\n--- AASHKA ---")
a = int(input("Enter Aashka's private key (a): "))
```

```
print("\n--- DEESHA ---")
b = int(input("Enter Deesha's private key (b): "))
```

```
A = pow(g, a, p)
```

```
B = pow(g, b, p)
```

```
Aashka_shared_key = pow(B, a, p)
Deesha_shared_key = pow(A, b, p)
```

```
print("\n" + "=" * 50)
print("    KEY EXCHANGE RESULTS")
print("=" * 50)
```

```
print("\nPublic Parameters:")
print("Prime number (p) :", p)
print("Generator (g)  :", g)
```

```
print("\nAashka:")
print("Private Key    :", a)
print("Public Key     :", A)
```

```
print("\nDeesha:")
print("Private Key    :", b)
print("Public Key     :", B)
```

NAME: AASHKA MISHRA

ROLL NO: T095

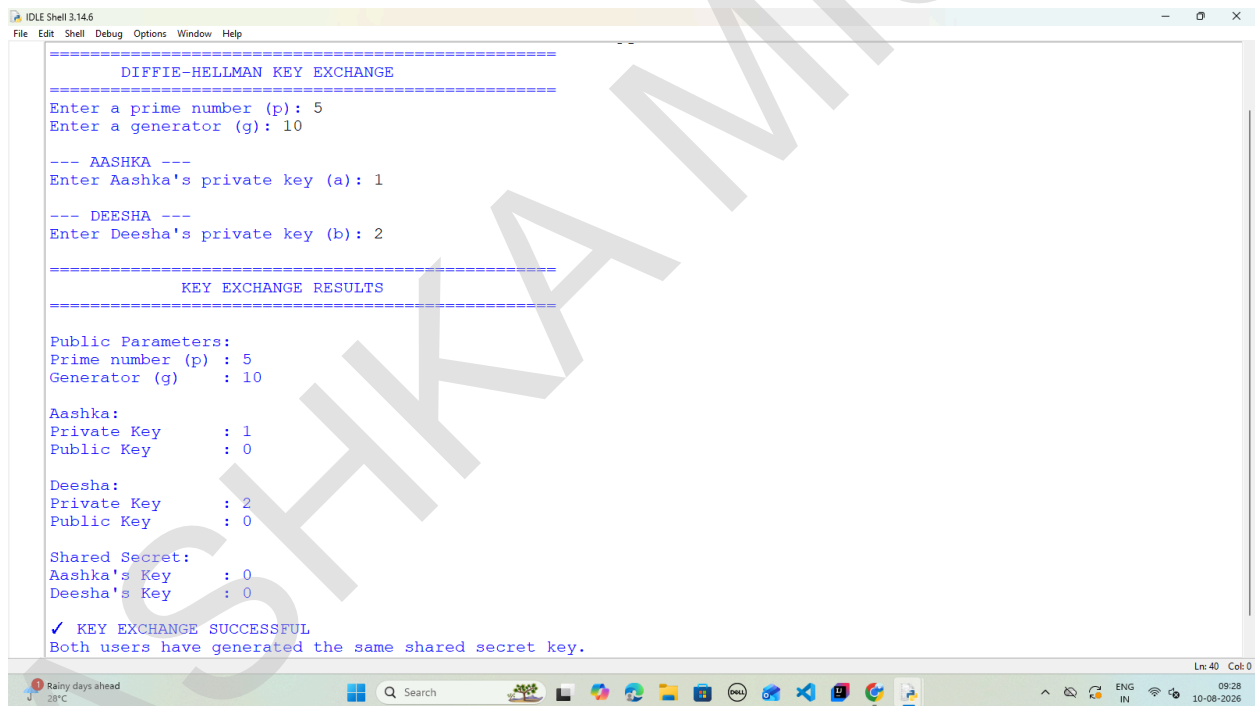
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```
print("\nShared Secret:")
print("Aashka's Key   :", Aashka_shared_key)
print("Deesha's Key   :", Deesha_shared_key)

if Aashka_shared_key == Deesha_shared_key:
    print("\n✓ KEY EXCHANGE SUCCESSFUL")
    print("Both users have generated the same shared secret key.")
else:
    print("\n✗ KEY EXCHANGE FAILED")
    print("The shared keys do not match.")

print("=" * 50)
```

OUTPUT:



```
=====
DIFFIE-HELLMAN KEY EXCHANGE
=====
Enter a prime number (p): 5
Enter a generator (g): 10

--- AASHKA ---
Enter Aashka's private key (a): 1

--- DEESHA ---
Enter Deesha's private key (b): 2

=====
KEY EXCHANGE RESULTS
=====

Public Parameters:
Prime number (p) : 5
Generator (g)   : 10

Aashka:
Private Key      : 1
Public Key       : 0

Deesha:
Private Key      : 2
Public Key       : 0

Shared Secret:
Aashka's Key     : 0
Deesha's Key     : 0

✓ KEY EXCHANGE SUCCESSFUL
Both users have generated the same shared secret key.

Ln: 40 Col: 0
```

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GUI:

```
import tkinter as tk
from tkinter import messagebox
import random

def calculate():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(aashka_private_entry.get())
        b = int(deesha_private_entry.get())

        if p <= 1:
            messagebox.showerror(
                "Invalid Input",
                "Prime number p must be greater than 1."
            )
            return

        if g <= 0:
            messagebox.showerror(
                "Invalid Input",
                "Generator g must be greater than 0."
            )
            return

        if a <= 0 or b <= 0:
            messagebox.showerror(
                "Invalid Input",
                "Private keys must be positive numbers."
            )
            return

        if g >= p:
            messagebox.showerror(
                "Invalid Input",
                "Generator g should be smaller than p."
            )
            return
```

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```
aashka_public = pow(g, a, p)
```

```
deesha_public = pow(g, b, p)
```

```
aashka_shared = pow(deesha_public, a, p)
```

```
deesha_shared = pow(aashka_public, b, p)
```

```
aashka_public_value.config(  
    text=str(aashka_public),  
    fg="#5EE7DF"  
)
```

```
deesha_public_value.config(  
    text=str(deesha_public),  
    fg="#A78BFA"  
)
```

```
aashka_shared_value.config(  
    text=str(aashka_shared),  
    fg="#A7F3D0"  
)
```

```
deesha_shared_value.config(  
    text=str(deesha_shared),  
    fg="#A7F3D0"  
)
```

```
calculation_text.delete(  
    "1.0",  
    tk.END  
)
```

```
calculation_text.insert(  
    tk.END,  
    "DIFFIE-HELLMAN KEY EXCHANGE PROCESS\n"
```

```
"=====\\n\\n"
```

```
)
```

```
calculation_text.insert(
    tk.END,
    "STEP 1 — PUBLIC PARAMETERS\n\n"
)

calculation_text.insert(
    tk.END,
    f"Prime number (p) = {p}\n"
    f"Generator (g) = {g}\n\n"
)

calculation_text.insert(
    tk.END,
    "STEP 2 — AASHKA GENERATES PUBLIC KEY\n\n"
)

calculation_text.insert(
    tk.END,
    f"A = g^a mod p\n"
    f"A = {g}^{a} mod {p}\n"
    f"A = {aashka_public}\n\n"
)

calculation_text.insert(
    tk.END,
    "STEP 3 — DEESHA GENERATES PUBLIC KEY\n\n"
)

calculation_text.insert(
    tk.END,
    f"B = g^b mod p\n"
    f"B = {g}^{b} mod {p}\n"
    f"B = {deesha_public}\n\n"
)

calculation_text.insert(
    tk.END,
    "STEP 4 — AASHKA CALCULATES SHARED SECRET\n\n"
)

calculation_text.insert(
```

```
tk.END,  
f"Ka = Ba mod p\n"  
f"Ka = {deesha_public}^{a} mod {p}\n"  
f"Ka = {aashka_shared}\n\n"  
)  
  
calculation_text.insert(  
    tk.END,  
    "STEP 5 — DEESHA CALCULATES SHARED SECRET\n\n"  
)  
  
calculation_text.insert(  
    tk.END,  
    f"Kβ = Ab mod p\n"  
    f"Kβ = {aashka_public}^{b} mod {p}\n"  
    f"Kβ = {deesha_shared}\n\n"  
)  
  
calculation_text.insert(  
    tk.END,  
    "STEP 6 — VERIFY SHARED KEY\n\n"  
)  
  
if aashka_shared == deesha_shared:  
  
    calculation_text.insert(  
        tk.END,  
        "✓ SUCCESS!\n\n"  
        "Aashka's shared key = Deesha's shared key\n"  
        f"Shared Secret Key = {aashka_shared}\n\n"  
        "The Diffie–Hellman key exchange was successful."  
    )  
  
    status_label.config(  
        text="● KEY EXCHANGE SUCCESSFUL",  
        fg="#4ADE80"  
    )  
  
    status_box.config(  
        bg="#123C2C"
```

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```
)

else:

    calculation_text.insert(
        tk.END,
        "✗ ERROR!\n\n"
        "The shared keys do not match."
    )

    status_label.config(
        text="● KEY EXCHANGE FAILED",
        fg="#FB7185"
    )

    status_box.config(
        bg="#421B24"
    )

except ValueError:

    messagebox.showerror(
        "Invalid Input",
        "Please enter valid integer values."
    )

def reset_all():

    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)

    aashka_private_entry.delete(0, tk.END)
    deesha_private_entry.delete(0, tk.END)

    p_entry.insert(0, "23")
    g_entry.insert(0, "5")

    aashka_private_entry.insert(0, "6")
    deesha_private_entry.insert(0, "15")
```

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```
aashka_public_value.config(  
    text="—",  
    fg="#5EE7DF"  
)  
  
deesha_public_value.config(  
    text="—",  
    fg="#A78BFA"  
)  
  
aashka_shared_value.config(  
    text="—",  
    fg="#A7F3D0"  
)  
  
deesha_shared_value.config(  
    text="—",  
    fg="#A7F3D0"  
)  
  
calculation_text.delete(  
    "1.0",  
    tk.END  
)  
  
calculation_text.insert(  
    tk.END,  
    "Enter the parameters and click\n"  
    "\"GENERATE SHARED KEY\" to begin.\n\n"  
    "The complete Diffie–Hellman\n"  
    "calculation will appear here."  
)  
  
status_label.config(  
    text="● WAITING FOR KEY EXCHANGE",  
    fg="#FBBF24"  
)  
  
status_box.config(  
    bg="#332A12"
```


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)

```
def generate_random_keys():
```

```
    try:
```

```
        p = int(p_entry.get())
```

```
        if p <= 10:
```

```
            messagebox.showwarning(
                "Prime Number",
                "Enter a larger value of p first."
            )
```

```
        return
```

```
        aashka_private_entry.delete(
            0,
            tk.END
        )
```

```
        deesha_private_entry.delete(
            0,
            tk.END
        )
```

```
        aashka_private_entry.insert(
            0,
            str(random.randint(2, p - 2))
        )
```

```
        deesha_private_entry.insert(
            0,
            str(random.randint(2, p - 2))
        )
```

```
    except ValueError:
        messagebox.showerror(
            "Invalid Input",
```

"Please enter a valid value for p."

)

root = tk.Tk()

root.title(

"Diffie–Hellman Key Exchange | Aashka & Deesha"

)

root.geometry(

"1200x800"

)

root.minsize(

1050,

700

)

root.configure(

bg="#07111F"

)

BG = "#07111F"

CARD = "#0D1B2A"

CARD2 = "#102235"

BORDER = "#1E3A5F"

WHITE = "#F8FAFC"

TEXT = "#CBD5E1"

MUTED = "#7891AA"

CYAN = "#5EE7DF"

GREEN = "#4ADE80"

YELLOW = "#FBBF24"

RED = "#FB7185"

header = tk.Frame(

root,

bg=BG,

height=100

)

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```
header.pack(
    fill="x",
    padx=35,
    pady=(25, 0)
)

logo = tk.Label(
    header,
    text="🔒",
    font=("Segoe UI Emoji", 32),
    bg=BG,
    fg=CYAN
)

logo.pack(
    side="left",
    padx=(0, 15)
)

title_frame = tk.Frame(
    header,
    bg=BG
)

title_frame.pack(
    side="left"
)

title = tk.Label(
    title_frame,
    text="DIFFIE-HELLMAN",
    font=("Segoe UI", 26, "bold"),
    bg=BG,
    fg=WHITE
)

title.pack(
    anchor="w"
)
```

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```
subtitle = tk.Label(  
    title_frame,  
    text="SECURE KEY EXCHANGE SIMULATOR",  
    font=("Segoe UI", 10, "bold"),  
    bg=BG,  
    fg=CYAN  
)
```

```
subtitle.pack(  
    anchor="w",  
    pady=(2, 0)  
)
```

```
version = tk.Label(  
    header,  
    text="CRYPTOGRAPHY • PRACTICAL",  
    font=("Segoe UI", 9, "bold"),  
    bg=BG,  
    fg=MUTED  
)
```

```
version.pack(  
    side="right",  
    pady=10  
)
```

```
main = tk.Frame(  
    root,  
    bg=BG  
)
```

```
main.pack(  
    fill="both",  
    expand=True,  
    padx=35,  
    pady=15  
)
```

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```
left = tk.Frame(  
    main,  
    bg=BG,  
    width=470  
)  
  
left.pack(  
    side="left",  
    fill="y",  
    padx=(0, 15)  
)  
  
left.pack_propagate(False)  
  
param_card = tk.Frame(  
    left,  
    bg=CARD,  
    highlightbackground=BORDER,  
    highlightthickness=1  
)  
  
param_card.pack(  
    fill="x",  
    pady=(0, 15)  
)  
  
tk.Label(  
    param_card,  
    text=" PUBLIC PARAMETERS",  
    font=("Segoe UI", 12, "bold"),  
    bg=CARD,  
    fg=WHITE  
)  
.pack(  
    anchor="w",  
    padx=20,  
    pady=(18, 3)  
)
```

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```
tk.Label(  
    param_card,  
    text="These values can be exchanged publicly.",  
    font=("Segoe UI", 9),  
    bg=CARD,  
    fg=MUTED  
).pack(  
    anchor="w",  
    padx=20,  
    pady=(0, 15)  
)
```

```
def create_input(parent, label, default):
```

```
    frame = tk.Frame(  
        parent,  
        bg=CARD  
    )
```

```
    frame.pack(  
        fill="x",  
        padx=20,  
        pady=6  
    )
```

```
    tk.Label(  
        frame,  
        text=label,  
        font=("Segoe UI", 10, "bold"),  
        bg=CARD,  
        fg=TEXT,  
        width=22,  
        anchor="w"  
    ).pack(  
        side="left"  
    )
```

```
    entry = tk.Entry(  
        frame,  
        textvariable=entry_var,  
        font=entry_font,  
        bg=entry_bg,  
        fg=entry_fg,  
        width=entry_width,  
        borderwidth=entry_borderwidth,  
        insertbackground=entry_insertbackground,  
        insertcolor=entry_insertcolor,  
        validate=entry_validate,  
        validatecommand=entry_validatecommand,  
        validateafter=entry_validateafter,  
        show=entry_show,  
        state=entry_state,  
        style=entry_style,  
        takefocus=entry_takefocus,  
        text=entry_text,  
        textcolor=entry_textcolor,  
        textvariable=entry_textvariable,  
        width=entry_width,  
        wraplength=entry_wraplength,  
        wrapmode=entry_wrapmode,  
        wrap=entry_wrap,  
        x=entry_x,  
        x2=entry_x2,  
        y=entry_y,  
        y2=entry_y2,  
        z=entry_z,  
        z2=entry_z2,  
        z3=entry_z3,  
        z4=entry_z4,  
        z5=entry_z5,  
        z6=entry_z6,  
        z7=entry_z7,  
        z8=entry_z8,  
        z9=entry_z9,  
        z10=entry_z10,  
        z11=entry_z11,  
        z12=entry_z12,  
        z13=entry_z13,  
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        z15=entry_z15,  
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        z20=entry_z20,  
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        z36=entry_z36,  
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        z149=entry_z149,  
        z150=entry_z150,  
        z151=entry_z151,  
        z152=entry_z152,  
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        z221=entry_z221,  
        z222=entry_z222,  
        z223=entry_z223,  
        z224=entry_z224,  
        z225=entry_z225,  
        z226=entry_z226,  
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        z234=entry_z234,  
        z235=entry_z235,  
        z236=entry_z236,  
        z237=entry_z237,  
        z238=entry_z238,  
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```

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

```
frame,  
font=("Consolas", 11),  
bg="#081522",  
fg=WHITE,  
insertbackground=CYAN,  
relief="flat",  
highlightthickness=1,  
highlightbackground=BORDER,  
highlightcolor=CYAN  
)
```

```
entry.pack(  
    side="right",  
    fill="x",  
    expand=True,  
    ipady=7  
)
```

```
entry.insert(  
    0,  
    default  
)
```

```
return entry
```

```
p_entry = create_input(  
    param_card,  
    "Prime number (p)",  
    "23"  
)
```

```
g_entry = create_input(  
    param_card,  
    "Generator (g)",  
    "5"  
)
```

```
tk.Label(  

```

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

```
param_card,
text="",
bg=CARD
).pack(
    pady=3
)

users_card = tk.Frame(
    left,
    bg=CARD,
    highlightbackground=BORDER,
    highlightthickness=1
)

users_card.pack(
    fill="x",
    pady=(0, 15)
)

tk.Label(
    users_card,
    text=" PARTICIPANTS",
    font=("Segoe UI", 12, "bold"),
    bg=CARD,
    fg=WHITE
).pack(
    anchor="w",
    padx=20,
    pady=(18, 15)
)

aashka_frame = tk.Frame(
    users_card,
    bg="#0C2730"
)

aashka_frame.pack(
    fill="x",
    padx=15,
```

NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

```
pady=(0, 8)  
)
```

```
tk.Label(  
    aashka_frame,  
    text="👩",  
    font=("Segoe UI Emoji", 20),  
    bg="#0C2730"  
)  
.pack(  
    side="left",  
    padx=12  
)
```

```
aashka_info = tk.Frame(  
    aashka_frame,  
    bg="#0C2730"  
)
```

```
aashka_info.pack(  
    side="left",  
    fill="x",  
    expand=True  
)
```

```
tk.Label(  
    aashka_info,  
    text="AASHKA",  
    font=("Segoe UI", 10, "bold"),  
    bg="#0C2730",  
    fg="CYAN"  
)  
.pack(  
    anchor="w",  
    pady=(7, 0)  
)
```

```
tk.Label(  
    aashka_info,  
    text="AASHKA",  
    font=("Segoe UI", 10, "bold"),  
    bg="#0C2730",  
    fg="CYAN"  
)  
.pack(  
    anchor="w",  
    pady=(7, 0)  
)
```

NAME: AASHKA MISHRA
ROLL NO: T095

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```
aashka_info,  
text="Private key (a)",  
font=("Segoe UI", 9),  
bg="#0C2730",  
fg=MUTED  
)  
.pack(  
    anchor="w"  
)  
  
aashka_private_entry = tk.Entry(  
    aashka_frame,  
    font=("Consolas", 11, "bold"),  
    width=8,  
    bg="#081522",  
    fg=WHITE,  
    insertbackground=CYAN,  
    relief="flat",  
    justify="center"  
)  
  
aashka_private_entry.pack(  
    side="right",  
    padx=15,  
    pady=5  
)  
  
aashka_private_entry.insert(  
    0,  
    "6"  
)  
  
deesha_frame = tk.Frame(  
    users_card,  
    bg="#171F35"  
)  
deesha_frame.pack(  
    fill="x",  
    padx=15,  
    pady=(0, 15)
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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)

```
tk.Label(  
    deesha_frame,  
    text="👤",  
    font=("Segoe UI Emoji", 20),  
    bg="#171F35"  
).pack(  
    side="left",  
    padx=12  
)
```

```
deesha_info = tk.Frame(  
    deesha_frame,  
    bg="#171F35"  
)
```

```
deesha_info.pack(  
    side="left",  
    fill="x",  
    expand=True  
)
```

```
tk.Label(  
    deesha_info,  
    text="DEESHA",  
    font=("Segoe UI", 10, "bold"),  
    bg="#171F35",  
    fg="#A78BFA"  
).pack(  
    anchor="w",  
    pady=(7, 0)  
)
```

```
tk.Label(  
    deesha_info,  
    text="Private key (b)",  
    font=("Segoe UI", 9),
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
bg="#171F35",  
fg=MUTED  
)  
.pack(  
    anchor="w"  
)
```

```
deesha_private_entry = tk.Entry(  
    deesha_frame,  
    font=("Consolas", 11, "bold"),  
    width=8,  
    bg="#081522",  
    fg=WHITE,  
    insertbackground=CYAN,  
    relief="flat",  
    justify="center"  
)
```

```
deesha_private_entry.pack(  
    side="right",  
    padx=15,  
    ipady=5  
)
```

```
deesha_private_entry.insert(  
    0,  
    "15"  
)
```

```
button_frame = tk.Frame(  
    left,  
    bg=BG  
)
```

```
button_frame.pack(  
    fill="x"  
)
```

```
generate_button = tk.Button(  
    button_frame,  
    text="⚡ GENERATE SHARED KEY",  
    command=calculate,
```

```
font=("Segoe UI", 11, "bold"),
bg="#0EA5A4",
fg="white",
activebackground="#14B8A6",
activeforeground="white",
relief="flat",
cursor="hand2",
bd=0
)

generate_button.pack(
    fill="x",
    ipady=11,
    pady=(0, 8)
)

random_button = tk.Button(
    button_frame,
    text="🔑 RANDOMIZE PRIVATE KEYS",
    command=generate_random_keys,
    font=("Segoe UI", 10, "bold"),
    bg=CARD2,
    fg=TEXT,
    activebackground=BORDER,
    activeforeground=WHITE,
    relief="flat",
    cursor="hand2",
    bd=0
)

random_button.pack(
    side="left",
    fill="x",
    expand=True,
    ipady=8,
    padx=(0, 5)
)
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
reset_button = tk.Button(  
    button_frame,  
    text="🔄 RESET",  
    command=reset_all,  
    font=("Segoe UI", 10, "bold"),  
    bg=CARD2,  
    fg=TEXT,  
    activebackground=BORDER,  
    activeforeground=WHITE,  
    relief="flat",  
    cursor="hand2",  
    bd=0  
)  
  
reset_button.pack(  
    side="right",  
    fill="x",  
    expand=True,  
    ipady=8,  
    padx=(5, 0)  
)  
  
right = tk.Frame(  
    main,  
    bg=BG  
)  
  
right.pack(  
    side="right",  
    fill="both",  
    expand=True  
)  
  
exchange_card = tk.Frame(  
    right,  
    bg=CARD,  
    highlightbackground=BORDER,  
    highlightthickness=1  
)
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
exchange_card.pack(  
    fill="x",  
    pady=(0, 15)  
)
```

```
tk.Label(  
    exchange_card,  
    text=" KEY EXCHANGE VISUALIZATION",  
    font=("Segoe UI", 12, "bold"),  
    bg=CARD,  
    fg=WHITE  
)  
.pack(  
    anchor="w",  
    padx=20,  
    pady=(15, 10)  
)
```

```
visual = tk.Frame(  
    exchange_card,  
    bg=CARD  
)
```

```
visual.pack(  
    fill="x",  
    padx=20,  
    pady=(0, 18)  
)
```

```
aashka_visual = tk.Frame(  
    visual,  
    bg="#0C2730",  
    width=150,  
    height=95  
)  
aashka_visual.pack(  
    side="left",  
    fill="both",  
    expand=True
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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)

aashka_visual.pack_propagate(False)

```
tk.Label(  
    aashka_visual,  
    text="👤",  
    font=("Segoe UI Emoji", 24),  
    bg="#0C2730"  
).pack(  
    pady=(7, 0)  
)
```

```
tk.Label(  
    aashka_visual,  
    text="AASHKA",  
    font=("Segoe UI", 10, "bold"),  
    bg="#0C2730",  
    fg="CYAN"  
).pack()
```

```
tk.Label(  
    visual,  
    text="—————▶\nPUBLIC KEY",  
    font=("Consolas", 8, "bold"),  
    bg=CARD,  
    fg="CYAN"  
).pack(  
    side="left",  
    padx=8  
)
```

```
middle_visual = tk.Frame(  
    visual,  
    bg="#101C2C",  
    width=150,  
    height=95  
)
```


SHETH L.U.J. AND SIR M.V. COLLEGE
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```
middle_visual.pack(  
    side="left",  
    fill="both",  
    expand=True  
)
```

```
middle_visual.pack_propagate(False)
```

```
tk.Label(  
    middle_visual,  
    text="🔑",  
    font=("Segoe UI Emoji", 24),  
    bg="#101C2C"  
)  
.pack(  
    pady=(7, 0)  
)
```

```
tk.Label(  
    middle_visual,  
    text="SHARED SECRET",  
    font=("Segoe UI", 9, "bold"),  
    bg="#101C2C",  
    fg=GREEN  
)  
.pack()
```

```
tk.Label(  
    visual,  
    text="PUBLIC KEY ← ",  
    font=("Consolas", 8, "bold"),  
    bg=CARD,  
    fg="#A78BFA"  
)  
.pack(  
    side="left",  
    padx=8  
)
```

```
deesha_visual = tk.Frame(  
    visual,  
    width=100,  
    height=100,  
    bg="white",  
    borderwidth=1,  
    border="solid",  
    bordercolor="black",  
    borderwidth=2  
)  
deesha_visual.pack()
```

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ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
visual,
bg="#171F35",
width=150,
height=95
)

deesha_visual.pack(
    side="left",
    fill="both",
    expand=True
)

deesha_visual.pack_propagate(False)

tk.Label(
    deesha_visual,
    text="👤",
    font=("Segoe UI Emoji", 24),
    bg="#171F35"
).pack(
    pady=(7, 0)
)

tk.Label(
    deesha_visual,
    text="DEESHA",
    font=("Segoe UI", 10, "bold"),
    bg="#171F35",
    fg="#A78BFA"
).pack()

result_frame = tk.Frame(
    right,
    bg=BG
)

result_frame.pack(
    fill="x",
```

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```
pady=(0, 15)
)

aashka_result = tk.Frame(
    result_frame,
    bg=CARD,
    highlightbackground=BORDER,
    highlightthickness=1
)

aashka_result.pack(
    side="left",
    fill="both",
    expand=True,
    padx=(0, 7)
)

tk.Label(
    aashka_result,
    text="AASHKA'S PUBLIC KEY",
    font=("Segoe UI", 9, "bold"),
    bg=CARD,
    fg=MUTED
).pack(
    pady=(12, 2)
)

aashka_public_value = tk.Label(
    aashka_result,
    text="—",
    font=("Consolas", 20, "bold"),
    bg=CARD,
    fg=CYAN
)
aashka_public_value.pack(
    pady=(0, 12)
)
```

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
deesha_result = tk.Frame(  
    result_frame,  
    bg=CARD,  
    highlightbackground=BORDER,  
    highlightthickness=1  
)
```

```
deesha_result.pack(  
    side="left",  
    fill="both",  
    expand=True,  
    padx=(7, 0)  
)
```

```
tk.Label(  
    deesha_result,  
    text="DEESHA'S PUBLIC KEY",  
    font=("Segoe UI", 9, "bold"),  
    bg=CARD,  
    fg=MUTED  
)  
.pack(  
    pady=(12, 2)  
)
```

```
deesha_public_value = tk.Label(  
    deesha_result,  
    text="—",  
    font=("Consolas", 20, "bold"),  
    bg=CARD,  
    fg="#A78BFA"  
)
```

```
deesha_public_value.pack(  
    pady=(0, 12)  
)
```

```
shared_card = tk.Frame(  
    right,
```

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```
        bg="#0B2520",
        highlightbackground="#1F5B4A",
        highlightthickness=1
    )

    shared_card.pack(
        fill="x",
        pady=(0, 15)
    )

    tk.Label(
        shared_card,
        text="🔑 SHARED SECRET KEY",
        font=("Segoe UI", 11, "bold"),
        bg="#0B2520",
        fg=GREEN
    ).pack(
        pady=(12, 5)
    )

    shared_values = tk.Frame(
        shared_card,
        bg="#0B2520"
    )

    shared_values.pack(
        fill="x",
        padx=25,
        pady=(0, 12)
    )

    aashka_shared_value = tk.Label(
        shared_values,
        text="—",
        font=("Consolas", 20, "bold"),
        bg="#0B2520",
        fg="#A7F3D0"
```

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: CYBER SECURITY

)

```
aashka_shared_value.pack(  
    side="left",  
    fill="x",  
    expand=True  
)
```

```
tk.Label(  
    shared_values,  
    text="=",  
    font=("Consolas", 18, "bold"),  
    bg="#0B2520",  
    fg=MUTED  
)  
.pack(  
    side="left"  
)
```

```
deesha_shared_value = tk.Label(  
    shared_values,  
    text="—",  
    font=("Consolas", 20, "bold"),  
    bg="#0B2520",  
    fg="#A7F3D0"  
)
```

```
deesha_shared_value.pack(  
    side="left",  
    fill="x",  
    expand=True  
)
```

```
calc_card = tk.Frame(  
    right,  
    bg=CARD,  
    highlightbackground=BORDER,  
    highlightthickness=1  
)
```

NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
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```
calc_card.pack(  
    fill="both",  
    expand=True  
)
```

```
tk.Label(  
    calc_card,  
    text=" STEP-BY-STEP CALCULATION",  
    font=("Segoe UI", 11, "bold"),  
    bg=CARD,  
    fg=WHITE  
)  
.pack(  
    anchor="w",  
    padx=20,  
    pady=(12, 7)  
)
```

```
calculation_text = tk.Text(  
    calc_card,  
    font=("Consolas", 9),  
    bg="#081522",  
    fg=TEXT,  
    insertbackground=CYAN,  
    relief="flat",  
    bd=0,  
    wrap="word"  
)
```

```
calculation_text.pack(  
    fill="both",  
    expand=True,  
    padx=15,  
    pady=(0, 15)  
)
```

```
calculation_text.insert(
```

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```
tk.END,  
"Enter the parameters and click\n"  
"\\"GENERATE SHARED KEY\\" to begin.\n\n"  
"The complete Diffie–Hellman\n"  
"calculation will appear here."  
)  
  
status_box = tk.Frame(  
    root,  
    bg="#332A12",  
    height=42  
)  
  
status_box.pack(  
    fill="x",  
    padx=35,  
    pady=(0, 20)  
)  
  
status_label = tk.Label(  
    status_box,  
    text="● WAITING FOR KEY EXCHANGE",  
    font=("Segoe UI", 9, "bold"),  
    bg="#332A12",  
    fg=YELLOW  
)  
  
status_label.pack(  
    pady=11  
)  
  
root.mainloop()
```


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OUTPUT:

The image displays two screenshots of a web-based "Diffie-Hellman Secure Key Exchange Simulator".

Top Screenshot: Shows the initial setup. Under "PUBLIC PARAMETERS", the Prime number (p) is 23 and the Generator (g) is 5. Under "PARTICIPANTS", AASHKA has a Private key (a) of 6 and DEESHA has a Private key (b) of 15. The "KEY EXCHANGE VISUALIZATION" section shows AASHKA's PUBLIC KEY as 8 and DEESHA'S PUBLIC KEY as 19. The "SHARED SECRET KEY" is shown as 2. The "STEP-BY-STEP CALCULATION" section shows the first three steps: STEP 1 - PUBLIC PARAMETERS (p=23, g=5), STEP 2 - AASHKA GENERATES PUBLIC KEY (A = g^a mod p = 5^6 mod 23 = 8), and STEP 3 - DEESHA GENERATES PUBLIC KEY (B = g^b mod p = 5^15 mod 23 = 19).

Bottom Screenshot: Shows the final steps of the process. The "SHARED SECRET KEY" remains 2. The "STEP-BY-STEP CALCULATION" section shows STEP 4 - AASHKA CALCULATES SHARED SECRET (K_s = B^a mod p = 19^6 mod 23 = 2), STEP 5 - DEESHA CALCULATES SHARED SECRET (K_s = A^b mod p = 8^15 mod 23 = 2), and STEP 6 - VERIFY SHARED KEY.

NAME: AASHKA MISHRA
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE
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Diffie-Hellman Key Exchange | Aashka & Deesha

DIFFIE-HELLMAN

SECURE KEY EXCHANGE SIMULATOR

CRYPTOGRAPHY • PRACTICAL

PUBLIC PARAMETERS
These values can be exchanged publicly.

Prime number (p): 23

Generator (g): 5

PARTICIPANTS

AASHKA
Private key (a): 6

DEESHA
Private key (b): 15

⚡ GENERATE SHARED KEY

RANDOMIZE PRIVATE KEYS RESET

KEY EXCHANGE VISUALIZATION

AASHKA'S PUBLIC KEY: 8

DEESHA'S PUBLIC KEY: 19

SHARED SECRET KEY: 2

STEP-BY-STEP CALCULATION

$K_A = 19^6 \bmod 23$
 $K_A = 2$

STEP 5 - DEESHA CALCULATES SHARED SECRET

$K_D = A^b \bmod p$
 $K_D = 8^{15} \bmod 23$
 $K_D = 2$

STEP 6 - VERIFY SHARED KEY

✓ SUCCESS!

Aashka's shared key = Deesha's shared key
Shared Secret Key = 2

The Diffie-Hellman key exchange was successful.

28°C Partly sunny

Search

ENG IN

09:20 10-08-2026

NAME: AASHKA MISHRA
ROLL NO: T095